

**SSD Anthology**

Anandtech's SSD Anthology is a great resource to get your basics of SSDs right  
<http://goo.gl/p0qz>

**USB flash drive manufacturing**

This video gives you a complete low down on how Kingston USB flash drives are made  
<http://goo.gl/S7mpI>

# Sleek Speed Demons

Solid state drives are sporting much better controllers, offering better write speeds at much better prices than what was seen a year back. We test 10 high end SSDs to help you decide which SSD you should have in your build



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**S**olid state drives have been around for a while now. With the pricing on hard drives showing no signs of reaching sensible levels, it is time you considered investing in an SSD. Of course, SSDs can still not match the cost per GB offered by HDDs. But if you want a system which is blazingly fast with quicker boot times, faster application loading times, etc., SSDs are the best bet.

Last year, we had conducted a mini SSD test in our Internal HDD test. This year we

decided that SSDs need to be tested separately, as they have come into their own as far as internal storage goes. Also, last year we had noticed that each SSD used a completely different SSD controller whereas all the SSDs we got for this test, sported the SandForce SF 2281 (except Corsair Force GT 240 GB which had SF 2282 controller) series controller.

SandForce has emerged as the most popular SSD controller off late for the great performance it offers. One of its main advantages is compressing data on the fly using the DuraWrite technology, which limits the number of times the flash memory is

written to – thereby increasing not only the performance, but also the lifetime of an SSD as there are fewer write cycles.

Another advantage of SandForce controller is that it does not require an external memory buffer. It features Redundant Array of Independent Silicon Elements (RAISE) technology to improve disk failure rates. Also SandForce controller sporting SSDs make use of some of the available flash memory for data parity, storing checksums, etc.

The next question on your mind must be - If the SSD controller is the same, wouldn't the performance of all SSDs be similar? The answer is no, as

the controller is just one part of an SSD, the other being the type of NAND memory used. The performance scores aren't similar if the memory type is synchronous or asynchronous.

Synchronous flash memory is more expensive as it can transfer data at both the rising and falling of the signal wave using the central timing circuit. This memory is also denoted by ONFI 2.x NAND or Toggle mode NAND memory. With the asynchronous flash chips, the interface ONFI 1 has a maximum bandwidth of 50 MB/s, thereby, affecting real world performance drastically.

Another important point to consider is that a larger